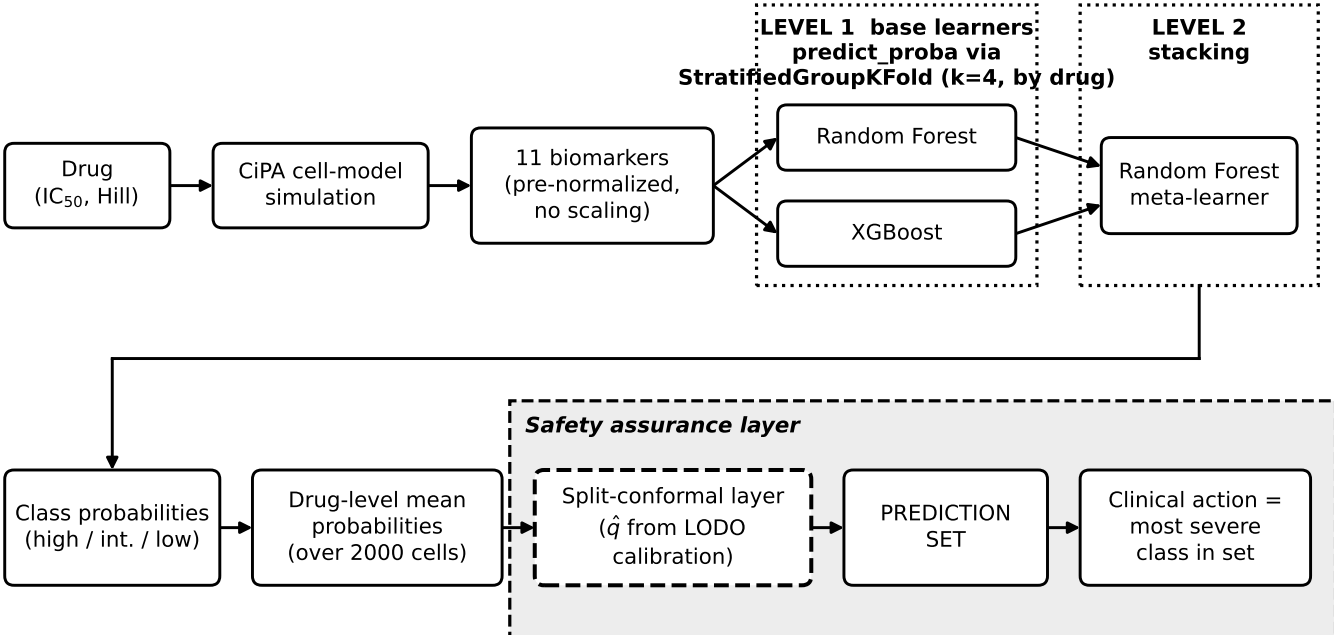
LEVEL 1 base learners
predict_proba via
StratifiedGroupKFold (k=4, by drug)
LEVEL 2
stacking
Drug
(IC_{50}, Hill)
CiPA cell-model
simulation
11 biomarkers
(pre-normalized,
no scaling)
Random Forest
XGBoost
Random Forest
meta-learner
Class probabilities
(high / int. / low)
Drug-level mean
probabilities
(over 2000 cells)
Safety assurance layer
Split-conformal layer
($\hat{q}$ from LODO
calibration)
PREDICTION
SET
Clinical action =
most severe
class in set